LAWS OF MOTION

INERTIA

A body cannot change its state of rest or uniform motion along a straight line. This property is called inertia.Inertia has no unit and no dimension.

INERTIA

- Inertia of Rest
- Inertia of Motion
- Inertia of Direction

- Inertia of Rest

Inability to change state of rest by itself.

- Inertia of Motion

Inability of a body to change its state of uniform motion by itself.

- Inertia of Direction

Inability of a body to change direction of motion by itself.

Newton's Second Law

F_{net} = Rate of change of linear momentum.

Instantaneous: $\vec{F} = \frac{d\vec{p}}{dt}$

Average: $\vec{F}_{av} = \frac{\Delta\vec{p}}{\Delta t} = \frac{\vec{p}_f - \vec{p}_i}{\Delta t}$

MOMENTUM

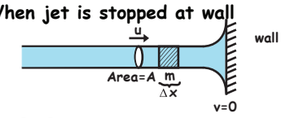
$\vec{P} = m\vec{v}$

-It is a vector quantity having direction same as that of velocity

-Unit is kg m/s.

LIQUID JETS

When jet is stopped at wall



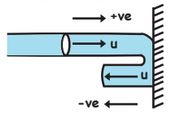
$F_{jet} = \frac{P_f - P_i}{\Delta t} = \frac{mx0 - mu}{\Delta t}$

$= \frac{-mu}{\Delta t} = -\rho A u^2 \left(m = \rho A \Delta x,\ \frac{\Delta x}{\Delta t} = u\right)$

$F_{wall} = \rho A u^2$

When liquid jet bounce back

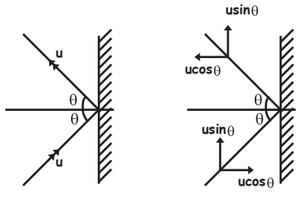
$F_{jet} = \frac{P_f - P_i}{\Delta t} = \frac{-2mu}{\Delta t}$



$F_{jet} = -2\rho A u^2$

$F_{wall} = 2\rho A u^2$

When liquid jet strikes obliquely



$F_{jet} = -2\rho A u^2 \cos\theta$ $F_{wall} = 2\rho A u^2 \cos\theta$

Change in momentum=-2mu cos θ

NEWTON'S THIRD LAW

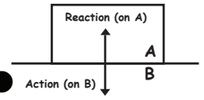
-To every action, there is always an equal (in magnitude) and opposite (in direction) reaction.

- Forces in nature always occur in pairs.
- A single isolated force is not possible.
- Counter force experienced by a body- reaction
- Action and reaction never act on the same body

* Force exerted on body A by body B (action)

* force exerted on body B by body A (reaction)

$\vec{F}_{Action} = -\vec{F}_{Reaction}$

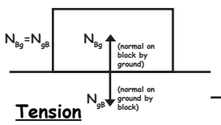


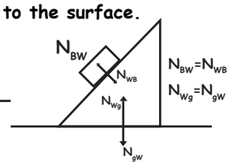
Some common forces

1.Normal Reaction

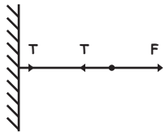
- Occurs when two surfaces are in contact with each other

Always perpendicular to the surface.



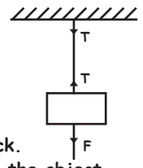


Tension



Restoring force developed when a longitudnal force is applied on a body

Ideal Rope



*Massless

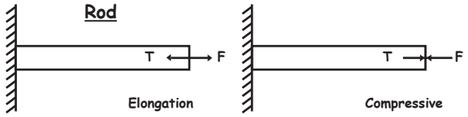
*Tension is same everywhere

*Opposes only elongation

*On compression it becomes slack.

*Tension always acts away from the object.

Rod



can support both elongation and compression

SINGLE BLOCK

Horizontal Force

Acceleration is along x-axis only

Along y-axis $a_y=0$,

$N - Mg = Ma_y$

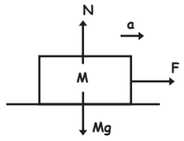
$\because a_y=0$

$N = Mg$

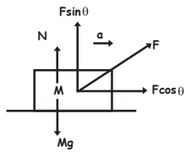
Along x-axis

$F - 0 = Ma$

$a = \frac{F}{M}$



Inclined Forces



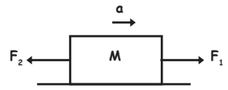
- If, Fsinθ <Mg
  block remains in contact with ground & accelerates horizontally
- If, Fsinθ =Mg
  block just leaves contact with ground
- If, Fsinθ >Mg
  the block leaves contact with ground and it begins to accelerate obliquily.

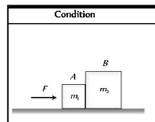
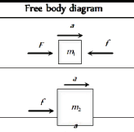
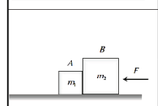
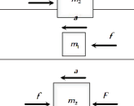
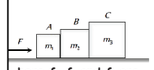
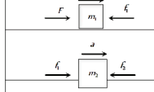
MOTION OF CONNECTED BODIES

$F_1 > F_2$

$F_1 - F_2 = Ma$

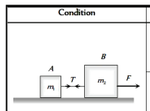
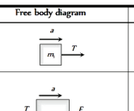
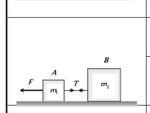
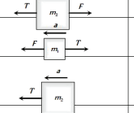
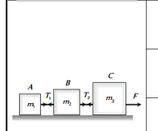
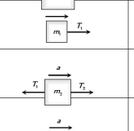
$a = \frac{F_1 - F_2}{M}$



Condition	Free body diagram	Equation	Force and acceleration
		$F - f = m_1 a$	$a = \frac{F}{m_1 + m_2}$
		$f = m_2 a$	$f = \frac{m_2 F}{m_1 + m_2}$
		$f = m_1 a$	$a = \frac{F}{m_1 + m_2}$
		$F - f = m_2 a$	$f = \frac{m_1 F}{m_1 + m_2}$
		$F - f_1 = m_1 a$	$a = \frac{F}{m_1 + m_2 + m_3}$
		$f_1 - f_2 = m_2 a$	$f_1 = \frac{(m_2 + m_3)F}{m_1 + m_2 + m_3}$
		$f_2 = m_3 a$	$f_2 = \frac{m_3 F}{m_1 + m_2 + m_3}$

here,f, f_1 and f_2 are normal reactions between blocks

MOTION OF BLOCKS CONNECTED BY MASSLESS STRING

Condition	Free body diagram	Equation	Tension and acceleration
		$T = m_1 a$	$a = \frac{F}{m_1 + m_2}$
		$F - T = m_2 a$	$T = \frac{m_1 F}{m_1 + m_2}$
		$F - T = m_1 a$	$a = \frac{F}{m_1 + m_2}$
		$T = m_2 a$	$T = \frac{m_2 F}{m_1 + m_2}$
		$T_1 = m_1 a$	$a = \frac{F}{m_1 + m_2 + m_3}$
		$T_2 - T_1 = m_2 a$	$T_1 = \frac{m_1 F}{m_1 + m_2 + m_3}$
		$F - T_2 = m_3 a$	$T_2 = \frac{(m_1 + m_2)F}{m_1 + m_2 + m_3}$

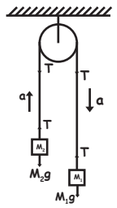
PULLEY-BLOCK SYSTEM

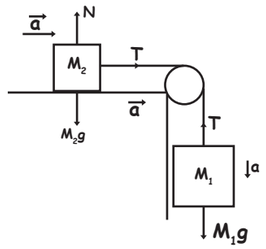
Ideal pulley

$M_1 > M_2$

$a = \frac{F}{M} = \frac{M_1 - M_2}{M_1 + M_2} g$

$T = \frac{2M_1M_2}{M_1 + M_2} g$

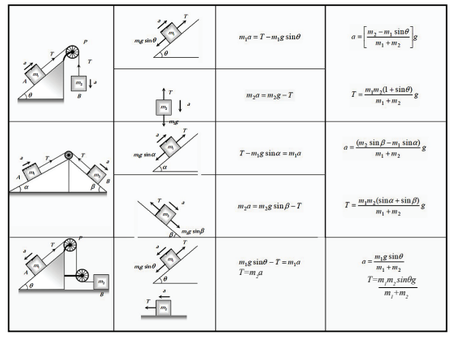




$a = \frac{M_1 g}{M_1 + M_2}$

$T = \frac{M_1 M_2}{M_1 + M_2} g$

INCLINED PLANE + PULLEY



THICK ROPE

Tension will be different at different points.

Mass per unit length= $\frac{M}{L}$

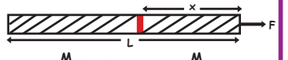
Mass of x length of rope = $\frac{M}{L}x$

Note :

Mass of given length= $\frac{\text{total mass}}{\text{total length}} \times$ given length

$\frac{\text{mass}}{\text{length}}$ = constant

$a_{rope} = \frac{F}{M}$



$\frac{M}{L}(L-x) = m_2$ $\frac{M}{L}x = m_1$

For(L-x) rope length,

$\vec{F} = m\vec{a} \Rightarrow T = m_2 \frac{F}{M}$

LIFT PROBLEMS

Apparent weight of body in a lift

Reading of weighing machine = reaction force exerted b weighing machine

Apparent weight ,($W_{apparent}$) = Reaction force (R)

Case 1: Lift is at rest

$R = mg$, $W_{apparent} = W_{actual} = mg$

Case 2: Lift moving up or down with constant velocity

$R = mg$, $W_{apparent} = W_{actual} = mg$

Case 3: Accelerated upward at a rate of 'a'

$R - mg = ma \Rightarrow R = m(g+a) = W_{app}$

$W_{apparent} > W_{actual} \rightarrow$ Feels over weight

Accelerated upward at a rate of 'g'

$R - mg = mg$, $R = 2mg$, $W_{app} = 2 \times W_{act}$

Case 4: Accelerated downward at a rate of 'a'

$mg - R = ma$, $R = m(g-a) = W_{app}$, $W_{app} < W_{act}$

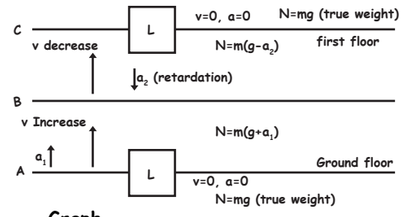
Accelerated downward at a rate of 'g' [Freefall]

$mg - R = mg$, $R = mg - mg = 0$, $W_{app} = 0$

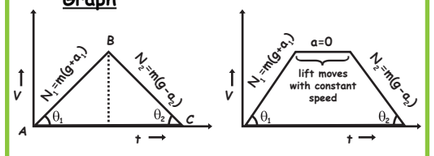
If a > g :

body leaves contact from ground and begins free-fall

Lift moves from ground floor to first floor



Graph



Acceleration,a_1=tanθ_1

Retardation,a_2=tanθ_2

FRAME OF REFERENCE & PSEUDO FORCE

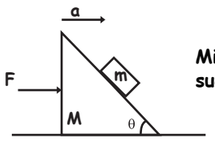
Frame of Reference

A frame in which observer is situated and makes his observation

Inertial frame of reference	Non-Inertial frame of reference
At rest or moving with uniform velocity along straight line. i.e unaccelerated	Accelerated frame of reference.
Newton's law of motion hold's $\vec{F}_{net} = m\vec{a}$	Newton's law of motion not applicable. $\vec{F}_{net} + \vec{F}_{pseudo} = m\vec{a}$ $\vec{F}_{pseudo} = -m\vec{a}_o$

LAWS OF MOTION

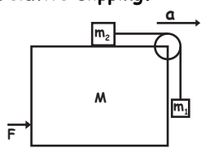
RELATIVE SLIPPING



Minimum force required to push the inclined plane such that "m" does not slip with respect to "M"

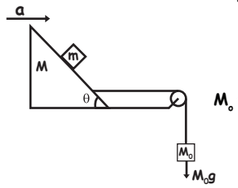
$F = (m+M)\, g\tan\theta$, $a = g\tan\theta$

Minimum acceleration of "M" such that there is no relative slipping.



$a = \left(\frac{m_1}{m_2}\right) g$

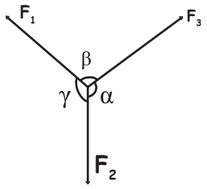
Minimum mass M_o such that there is no relative slipping



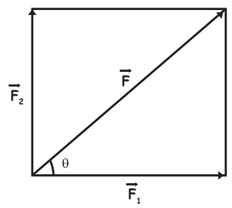
$M_o = \frac{M+m}{\cot\theta - 1}$

EQUILLIBRIUM & LAMI'S THEOREM

$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$



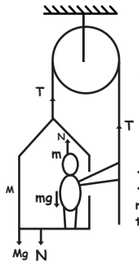
PARALLELOGRAM LAW



$\vec{F} = \vec{F_1} + \vec{F_2}$, $F = \sqrt{F_1^2 + F_2^2}$, $\tan\theta = \frac{F_2}{F_1}$

MAN-CAGE PROBLEM

Man holds the cage stationary



$T = \left(\frac{m+M}{2}\right) g$

$N = \left(\frac{m-M}{2}\right) g$

- m<M
- if m>M man cannot hold the cage stationary

IMPULSE

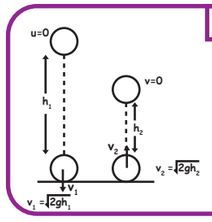
If a large force acting for short period of time, there will be a sudden change in momentum

$\vec{F}_{imp} = \frac{d\vec{p}}{dt}$

$\vec{P}_f - \vec{P}_i = \int_0^t \vec{F}_{imp}\, dt$ = area under F-t graph

$\vec{I}$ = Impulse = $\vec{P}_f - \vec{P}_i = \int_0^t \vec{F}_{imp}\, dt$ = area of F-t graph

Case-1



Impulse

$I = P_f - P_i$

$= mV_2 - (-mV_1)$

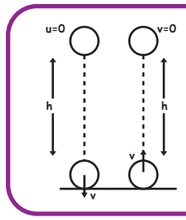
$I = m(V_1 + V_2)$

Average impulsive force

$F_{avg} = \frac{\text{impulse}}{t_o} = \frac{\Delta P}{t_o}$

$= \frac{m(v_1+v_2)}{t_o}$

Case-2



Impulse

$m(v+v) = J$

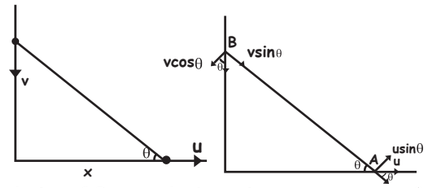
$= m(2v)$

$J = 2mv$

Average impulsive force

$F_{avg} = \frac{\Delta P}{\Delta t} = \frac{2mv}{t_o}$

ROD SLIDING ON A WALL



velocity of B towards A = vsin θ

velocity of A away from B = ucosθ

these velocities should be equal,

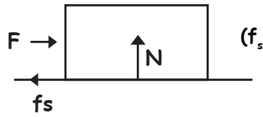
⇒ vsinθ = ucosθ

⇒ v = ucotθ

FRICTION

Static friction

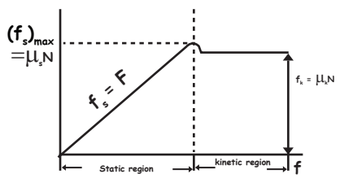
- It is a self adjusting force.
- The opposing force that comes into play, when object tends to slip over the surface of other object, but slipping has not yet started.
- As applied force increases static friction also increases.
- The body doesn't move until a maximum value of static friction is attained
- This value is called limiting friction or $(f_s)_{max}$



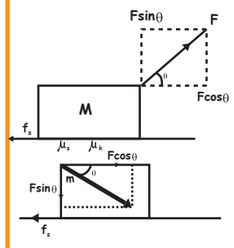
$(f_s)_{max} = \mu_s N$

Kinetic friction

If the applied force is increased further and slipping between surfaces start, the friction opposing the slipping is called kinetic friction.



PULLING FORCE & PUSHING FORCE



$F\cos\theta \geq \mu_s(mg - F\sin\theta)$

$F \geq \frac{\mu_s mg}{\cos\theta + \mu_s\sin\theta}$

$F_{pulling} = \frac{\mu_s mg}{\cos\theta + \mu_s\sin\theta}$

$F\cos\theta \geq \mu_s(mg + F\sin\theta)$

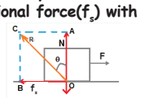
$F \geq \frac{\mu_s mg}{\cos\theta - \mu_s\sin\theta}$

$F_{pushing} = \frac{\mu_s mg}{\cos\theta - \mu_s\sin\theta}$

$F_{pushing} > F_{pulling}$

ANGLE OF FRICTION

Angle(θ) made by resultant of normal (N) & frictional force(f_s) with normal



$R = \sqrt{N^2 + f_s^2}$ R is resultant

$\tan\theta = \frac{f_s}{N}$

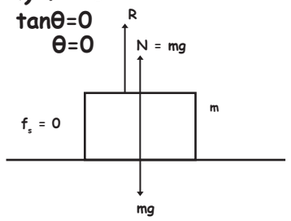
When $f_s = (f_s)_{max}$ $R = N\sqrt{1+\mu_s^2}$

$\tan\theta = \mu_s$, μ_s is Coefficient of friction

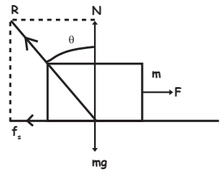
i) f = 0

tanθ=0

θ=0



ii) $0 < f_s < (f_s)_{max}$

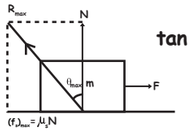


$0 \leq \tan\theta \leq \mu_s$

$N < R < N\sqrt{1+\mu_s^2}$

iii) $f_s = (f_s)_{max}$

$R_{max} = N\sqrt{1+\mu_s}$

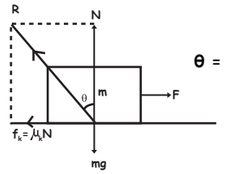


$\tan\theta_{max} = \mu_s$

iv) When slipping has started,

$f = f_k$

$R_{max} = N\sqrt{1+\mu_k^2}$



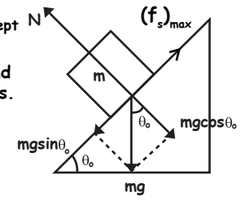
θ = Angle of kinetic friction

ANGLE OF REPOSE

Angle made by inclined plane such that a block kept on it just begins to slide

Depends only on μ_s and is independent of mass.

$\tan\theta_0 = \mu_s$



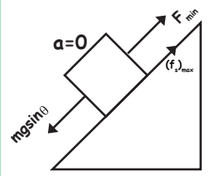
Variation of R

As angle of inclined plane increases, R remains constant (=mg) and when sliding starts R starts decreasing.

Variation of angle of friction.

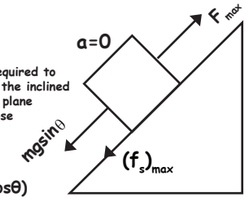
As angle of inclined plane increases, angle of friction will also increase and as sliding starts its value becomes constant and $\tan\theta = \mu_k$

Minimum & Maximum force (applied parallel to inclined plane)



$F_{min} = mg(\sin\theta - \mu_s\cos\theta)$

F_{min} is the minimum force required to keep the block stationary if the angle of inclined plane is greater than angle of repose



F_{max} is the maximum force required to keep the block stationary on the inclined plane if the angle of inclined plane is greater than angle of repose

$F_{max} = mg(\sin\theta + \mu_s\cos\theta)$

$mg(\sin\theta - \mu_s\cos\theta) \leq F \leq mg(\sin\theta + \mu_s\cos\theta)$

HORIZONTAL TRUCK BOX

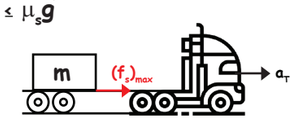
Case-1

Box does not slip.

$f_s \leq \mu_s N$

$\Rightarrow ma_T \leq \mu_s mg$

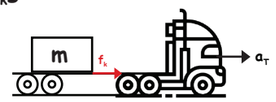
$\Rightarrow a_T \leq \mu_s g$



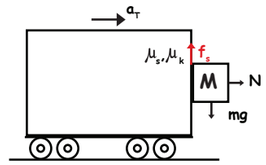
Case-2

Box slips

$a_T \geq \mu_k g$



VERTICAL TRUCK BOX



Case-1

Box does not slip.

$f_s \leq \mu_s N$

$\Rightarrow mg \leq \mu_s ma_T$

$\Rightarrow \mu_s \geq \frac{g}{a_T}$

Case-2

Box slips

$\mu_k \leq \frac{g}{a_T}$